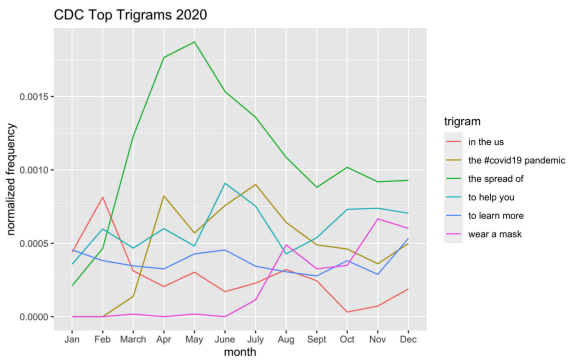
NHS, CDC, and WHO Twitter Health Communication: A Preliminary Shiny App

Ireland, Katherine
katherine.ireland@uga.edu
University of Georgia, United States of America

This work reports on ongoing and preliminary development of an R Shiny application to display and understand trends in health communication by three key public health agencies during 2020. Specifically, the data used for this application are a corpus of tweets by official accounts of the National Health Service (NHS), the Centers for Disease Control and Prevention (CDC), and the World Health Organization (WHO) throughout 2020. The dataset has been examined previously using different techniques through semantic analysis of trigrams and keywords (Ireland 2024) but has not been made available for users to interact with and analyze the texts. Figure 1 displays an example visualization of CDC trigrams for users to view.

Figure 1: CDC Top Trigrams 2020



At present, the application only includes English language tweets, although the full dataset includes other languages like Spanish and Hindi. Linguistic data and characteristics are gathered and compiled using R, specifically the following packages: tidytext (Silge and Robinson 2016), udpipe (Wijffels 2021), tidyverse (Wickham et al. 2019) and quanteda (Benoit et al. 2018). Visualizations are created using ggplot2 from the tidyverse (Wickham and Grolemund 2017, Wickham et al. 2019).

Related studies have considered the language of COVID-19 and of other health communication scenarios through news and social media, for example (Collins and Nerlich 2016, Gui et al. 2018, Hunt and Harvey 2015, Hunt and Brookes 2020, Semino and Koller 2020, Ireland 2024), but no studies have used R shiny to visualize textual analytics and differences in units like the three listed above. Additionally, the dataset provides a unique perspective on CDC, NHS, and WHO public health communication during the COVID-19 pandemic. Official accounts analyzed are available by request and are composed of over 2 million words.

This presentation will highlight the current version of the Shiny application and visualizations and discuss additional plans for development. This work will be of interest to interdisciplinary researchers because of the methodologies involved and because of the topic area of public and governmental health communication on social media.

Bibliography

Arnold, T . (2021). A tidy data model for natural 393 language processing using cleannlp. *IW3C2 (International World Wide Web Conference Committee).*

Collins, L., and Nerlich, B . (2016): “Uncertainty discourses in the context of climate change: A corpus-assisted analysis of UK national newspaper articles”, *Communications.* De Gruyter Mouton. (pp. 291-313).

Feinerer, I. (2020): “Introduction to the tm Package: Text Mining in R”, https://cran.r-project.org/web/packages/tm/vignettes/tm.pdf

Gabrielatos, C . (2018): “Keyness Analysis: Nature, Metrics and Techniques”, in Taylor, C / Marchi, A. (eds.): *Corpus Approaches To Discourse: A Critical Review* . Routledge.

Gui, X., Wang, Y., Kou, Y., Reynolds, T. L., Chen, Y., Mei, Q., and Zheng, K . (2018): “Understanding the Patterns of Health Information Dissemination on Social Media during the Zika Outbreak”, AMIA ... Annual Symposium proceedings. AMIA Symposium, 2017, 820–829.

Hunt, D., and Harvey, K. (2015): “Health Communication and Corpus Linguistics: Using Corpus Tools to Analyse Eating Disorder Discourse Online”, In: *Corpora and Discourse Studies* . Palgrave Advances in Language and Linguistics.

Hunt, D., and Brookes, G. (2020): *Corpus, Discourse and Mental Health* . Bloomsbury Publishing.

Ireland, K . (2024): “Wash your hands: CDC, WHO, and NHS tweets in the #COVID19 pandemic”, *Applied Corpus Linguistics.*

Python . Version 3.7.4. July 8 2019.

R Core Team . *(2014). R: A language and environment for statistical computing.* R Foundation for Statistical Computing. Vienna, Austria.

Semino, E., and Koller, V. (2020): “Talking about lockdown language”, Lancaster University Public Lecture.

Silge, J. and Robinson, D. (2016): *tidytext: Text mining and analysis using tidy data principles in r. JOSS, 1(3).*

Wickham, H., and Grolemund, G . (2017): *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data.* 1st ed. O’Reilly Media, Inc.

Wickham et al. (2019). “Welcome to the tidyverse”, *Journal of Open Source Software* , 4(43), 1686, https://doi.org/10.21105/joss.01686

WHO. (2009): “WHO: Public Health Campaigns: getting the message across”, WHO Press: Switzerland. **WHO.** (2021): “WHO Coronavirus (COVID-19) Dashboard”, World Health Organization. covid19.who.int

Wijffels, J. et al. (2023): “Udpipe: Tokenization, Parts of Speech Tagging, Lemmatization and Dependency Parsing with the ‘UDPipe’ ‘NLP’ Toolkit”. Version 0.8.11.

Zarcadoolas, C. (2010): *The simplicity complex: exploring simplified health messages in a complex world. Health Promotion International* . Oxford University Press.